Lesson Notes

MODULE 1 | LESSON 3

SYSTEMIC RISK: CONDITIONAL AND ILLIQUIDITY RISK

Reading Time	60 minutes
Prior Knowledge	Familiarity with systemic risk from prior two lessons in this module.
Keywords	Conditional Value at Risk, Distressed Insurance Premium, Co-Risk, Marginal and systemic expected shortfall, Illiquidity, Crowded trades

In the last lesson, we discussed macroeconomic measures of systemic risk, as well as how to quantify and visualize systemic risk as a network graph. In this lesson, we learn about more measures to identify and quantify potential sources of risk.

1. Distressed Insurance Premium

You already know about expected shortfall, the expected loss or shortfall conditional on the loss exceeding some threshold, where we often specify the 95th or 99th or 99.9th percentile as the threshold. Distressed insurance premium (DIP) is also an expected shortfall measure, but instead of conditional on exceeding a predefined quantile, it is "conditional on systemic distress" (Bisias et al. 105).

As you will read in the main required reading for this lesson, the DIP model requires three main inputs:

1. A firm's probability of default, which can be backed out of the credit default swap spread (once an assumption is made about the loss given default).
2. An estimated correlation matrix for the firm's assets, or alternatively, an average correlation.
3. This distress threshold that conditions expected shortfall. It can be defined as the monetary value of the total losses across the financial sector. To keep things simpler, we will define it as the percentage of total credit assets that have defaulted. Bisias et al. use 15% as this threshold input, but we will experiment below with this level to strengthen our intuition.

The reading by Bisias et al. summarizes the calculations, so here let us focus on a few stylized examples as examined with the help of the Python implementation of the model from the frds library.

Let's start with a "sunshine scenario," a relatively benign scenario where the financial circumstances do not seem to suggest high risk. In terms of the above inputs 1 and 2, what would you suggest? In a benign, low-risk scenario, we would posit low probabilities of default (PDs). Now, 1% is not a necessarily low (or high) PD, but let's use it for our ten banks (representing the entire banking system in our simplified model) by initializing the following variable:

```
default_probabilities = np.array([0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01,
0.01])
```

What about asset correlations indicated by the correlation matrix for our sunshine scenario? To take advantage of the risk-reducing effects of diversification, we would rather have low positive correlations among our assets or even negative correlations, so that some price decreases are cancelled out by other price increases for a less negative and loss volatile total return.

Let's take this correlation matrix of 10 assets:

```
neg_corr = np.array([[ 1. , -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1],

  [-0.1, 1. , -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1],
```

```
[-0.1, -0.1, 1. , -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1],

[-0.1, -0.1, -0.1, 1. , -0.1, -0.1, -0.1, -0.1, -0.1, -0.1],

[-0.1, -0.1, -0.1, -0.1, 1. , -0.1, -0.1, -0.1, -0.1, -0.1],

[-0.1, -0.1, -0.1, -0.1, -0.1, 1. , -0.1, -0.1, -0.1, -0.1],

[-0.1, -0.1, -0.1, -0.1, -0.1, -0.1, 1. , -0.1, -0.1, -0.1],

[-0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, 1. , -0.1, -0.1],

[-0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, 1. , -0.1],

[-0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, -0.1, 1. ]])
```

One simple way to construct this matrix in Python uses the following code:

```
neg_corr = np.ones((10, 10)) * -0.1

One_pt1 = np.eye(10) * 1.1

neg_corr = neg_corr + One_pt1
```

And to ensure that the matrix is positive definite so that it can be a valid concentration matrix, this line of code will return True if all the eigenvalues are positive:

`np.all(np.linalg.eigvals(neg_corr) > 0)`

Now, assuming we have frds installed, we import the DIP library:

```
from frds.measures import distress_insurance_premium

import numpy as np
```

And finding the DIP value could hardly be more straightforward:

`distress_insurance_premium(default_probabilities, neg_corr, default_threshold=0.15)`

This function should return a value near 0.000275, indicating that for a 1,000,000 euro balance sheet, the risk-neutral expectation of portfolio credit losses that exceed 15% is about 275 Euro. In a risk-neutral world under distressed circumstances, the hypothetical cost of insuring a 1 milion Euro portfolio of the banks impacted by these credit losses (the insurance premium) would be 275 Euro.

If the concept of insurance premium is not familiar enough to you, recall that a long put can protect the value of a long stock position. In this way, buying a put is like buying insurance on the underlying stock. Now you could imagine that 275 Euros is the current price for a put option with a 1 million Euro strike. The portfolio was originally worth 1.15 million Euros, but it has decreased 15%, so the current price of the option reflects the market's view of the risk of additional decline in the portfolio's value.

Now let’s observe a more realistic scenario, where the assets are positively correlated instead of negatively correlated, but we’ll leave the PDs alone. We construct a banking system where all the banks have a 0.6 historical (Pearson) correlation:

```
pt6_corr = np.ones((10, 10)) * 0.6

zero_pt1 = np.eye(10) * 0.4

pt6_corr = pt6_corr + zero_pt1

pt6_corr
```

The above code returns the desired matrix:

```
array([[1. , 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6],

    [0.6, 1. , 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6],

     [0.6, 0.6, 1. , 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6],

     [0.6, 0.6, 0.6, 1. , 0.6, 0.6, 0.6, 0.6, 0.6, 0.6],

     [0.6, 0.6, 0.6, 0.6, 1. , 0.6, 0.6, 0.6, 0.6, 0.6],

     [0.6, 0.6, 0.6, 0.6, 0.6, 1. , 0.6, 0.6, 0.6, 0.6],

     [0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 1. , 0.6, 0.6, 0.6],

     [0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 1. , 0.6, 0.6],

     [0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 1. , 0.6],

     [0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 1. ]])
```

We run the same function with the new correlation matrix input:

```
distress_insurance_premium(default_probabilities, pt6_corr, default_threshold=0.15)
```

Predictably, it yields a much higher 0.0216, which is more than 75 times the previous result, or 21,640 Euros for a 1 million Euro portfolio.

Increasing the correlation across all the banks to 0.9 almost doubles the hypothetical insurance premium on a 1 million Euro portfolio to 42,550 Euro.

You can easily imagine how changing the PDs will impact this premium amount, but you should experiment for yourself to witness the non-linearity of the impact.

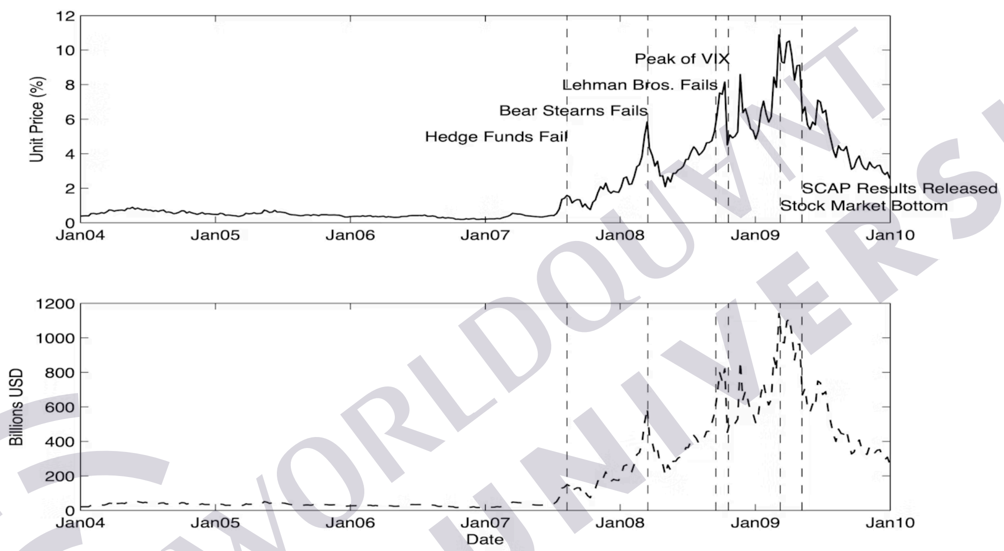
The direction of change in the premium value may not surprise you in the above examples. Perhaps you recall the non-linearity we observed in PD calculation from the Financial Markets course, in that case with respect to the recovery rate (= 1 – LGD), and so the non-linearity of the impact of the changing inputs to the DIP model may seem intuitive. Now ask yourself whether the premium value increases or decreases as a result of changing the above 15% threshold to 5%.

You might think that the DIP model would interpret a 15% impairment of the banking system assets as suggestive of a massive market dislocation that surely anticipates continued financial deterioration. This would imply that the cost of insuring against further losses should be higher for the 15% threshold than for the relatively less worrying 5% impairment. However, assuming that neither the correlation matrix nor the PDs have changed, the result of changing this threshold to 5% results in an increased insurance premium. There are two ways it might help to understand this. If we consider the threshold to be a deductible, as Huang et al. recommend (8), then the insurance premium increases as the threshold decreases because the hypothetical insurer must potentially pay out more and sooner: If the portfolio loses 10% of its value and the threshold is 5%, then the insurer must pay 5% of the initial portfolio value. If the threshold is 15%, however, then the insurer pays nothing because the 10% is less than the 15% “deductible” threshold. Another way to think about the same issue is to recall the put option parallel: A put option that is only 5% out of the money should be worth more than a put option that is 15% out of the money. Develop your intuition with respect to this systemic risk indicator by simply updating the `default_threshold` parameter to 0.05 and rerunning the scenarios we have discussed and/or some of your own.

If you were one of the people that assumed a higher impairment level should correspond to a higher systemic risk indicator value, remember that we did not make any changes to the model inputs. In a situation where 15% (or even 5%) of banks’ assets were written down as losses, the PDs implied by the credit default swap market would quickly reflect this, resulting in the higher hypothetical premium that you expected. Most likely, the forward-looking correlation matrix would also reflect higher correlations (tending closer to +1.0), which as we saw above also has the effect of increasing the hypothetical portfolio insurance premium. As Huang et al. emphasized on this point, “An increase in systemic risk related to concentration risk (measured by correlation) seems to have led to the onset of the 2007–09 financial crisis” (14).

To complement your understanding of DIP in order to implement it and similar measures (Monte Carlo simulations and all), you are encouraged to review the GitHub repo:
https://github.com/mgao6767/frds/blob/main/src/frds/measures/distress_insurance_premium.py

Figure 1: Historical Levels of the Distressed Insurance Premium through the Great Financial Crisis



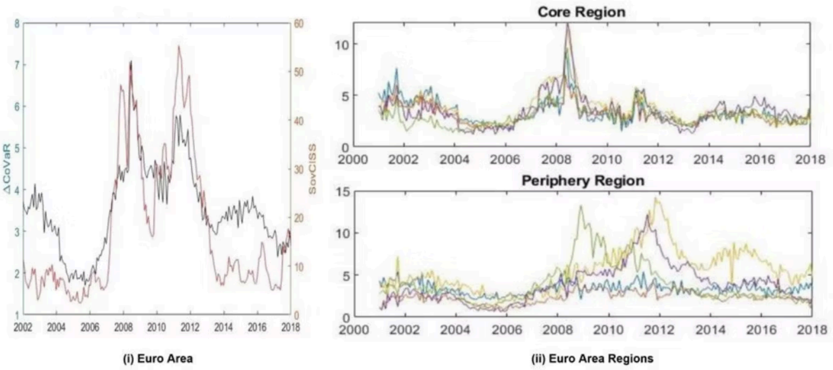
Source: Huang, Xin, et al. "Systemic Risk Contributions." *Federal Reserve Board Working Papers*, Finance and Economics Discussion Series, Aug 2011, https://www.federalreserve.gov/pubs/feds/2011/201108/201108pap.pdf.

2. Other Conditional Risk Measures

2.1 CoVaR

Just as DIP is the value of the hypothetical risk premium assuming a 15% loss across the banking system’s assets, CoRisk and CoVar measure risks conditioned on some other event. CoVar helps identify systemically risky institutions by showing how much the Value at Risk (VaR) of *the financial system at large changes* when we assume the firm in question is in distress: A large difference in the financial system’s VaR with and without the specific firm’s distress suggests that the firm is systemically important, i.e., it poses significant potential systemic risk. On the other hand, if the CoVaR level is basically the same regardless of whether we make this assumption, the specific firm does not contribute significantly to the potential systemic risk, according to this metric. As you can see in figure 2, the change in CoVaR represented by the black line on the left (i) does indeed suggest increasing systemic risk as early as 2006, in anticipation of the Great Financial Crisis starting in 2007–8. For comparison, the light red line is the Composite Indicator of Systemic Stress provided by the European Central Bank (Skouralis 6).

Figure 2: CoVaR Changes for 52 European Banks



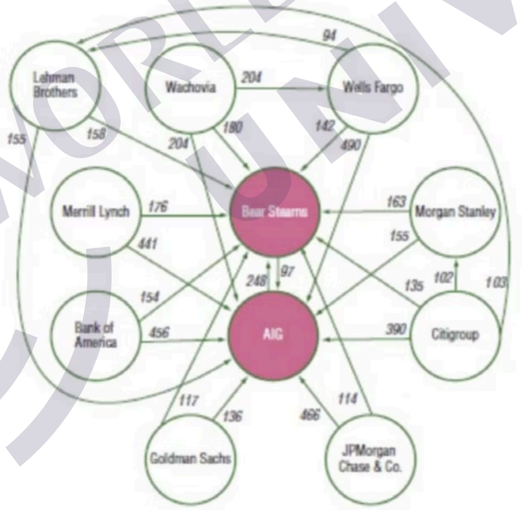
Source: Skouralis, Alexandros. "The Role of Systemic Risk Spillovers in the Transmission of Euro Area Monetary Policy." *European Systemic Risk Board*, Working Paper Series, no. 129, Dec. 2021, p. 6, https://www.esrb.europa.eu//pub/pdf/wp/esrb.wp129~01256b31ef.en.pdf.

2.2 CoRisk

CoRisk like CoVaR relies on quantile regression. Quantile regression is like the linear regression that you are most familiar with, except instead of calculating the effect of an independent variable on the mean value of a dependent variable, we estimate regression coefficients as the the independent variable's effect on a specified quantile (Clay). We should be thinking about systemic risk when CDS spreads for important financial institutions are at the high end of their range, when the spread is indicating higher risk of default. In the case of CoRisk, the relevant quantile is the 95th percentile of the CDS spread.

CoRisk predicts the percentage increase in other firms' credit spreads in response to (i.e., conditioned on) another particular firm's spread being at or above its own 95th percentile. When this analysis has been performed on all the most important participants in a financial market, we might find a network of potential impacts similar to the historical results shown in figure 3. Note how these impacts are directional and not symmetric: At the time, AIG's distress was estimated to increase the CDS spread of Bear Stearns by almost 250%, while Bear Stearns would *only* double AIG's spread. The emphasis on "only" is meant to highlight the irony. As you can see, many firms in the web of default risk below had the apparent potential to increase other firms' CDS spreads by even larger multiples.

Figure 3: CoRisk Estimates (Greater than or Equal to 90%) between Pairs of Selected Financial Institutions



Source: Shin, Hyun Song. "Financial Intermediation and the Post-Crisis Financial System." *Bank for International Settlements*, BIS Working Papers, Mar 2010, no. 304, https://www.bis.org/publ/work304.pdf.

3. More Python Implementations of Systemic Risk Measures

We mentioned the frds Python library above with reference to DIP, but this library actually includes implementations of several of the systemic risk measures discussed in the reading for these first three lessons. With these, you can easily experiment and improve your intuition about their relative sensitivities to varying inputs:

- Kyle's lambda
- Absorption ratio
- Contingent claim analysis
- Distressed insurance premium

- Marginal expected shortfall
- Systemic expected shortfall

The more you understand the mechanics at the level of the code, the better equipped you will be to adapt or further develop these measures for your future purposes. The repo for each of these is also available for your inspection: https://github.com/mgao6767/frds/tree/main/src/frds/measures

4. Conclusion

In this lesson, we completed our survey of the systemic risk measures. In the next lesson, we learn about how machine learning techniques are applied to the same problem of predicting and quantifying systemic risk. We take an especially close look at a particular study that builds on areas with which we are already specifically familiar: sentiment analysis (text mining), information theory, and Granger causality.

References

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